

METAL AND NON METALS

1. INTRODUCTION



Ravi, a class 8 student, hears about metals and non-metals for the first time. He wonders why some materials conduct electricity while others do not.

2. FINDING MATERIALS AROUND US

Ravi starts observing objects around him.

METALS



- Spoon (Steel)
- Aluminium vessel
- Copper wire

NON-METALS



- Plastic bottle
- Sulfur
- Pencil (wood)

3. CURIOSITY BEGINS

Ravi becomes curious and starts asking questions.



- Why are utensils made of metal?
- Why is plastic not used for electric wires?
- What makes metals different from non-metals?

4. DISCOVERING THE DIFFERENCES

Through experiments and classroom activities, Ravi learns the unique properties.

PROPERTY	METALS	NON-METALS
Appearance	Shiny (lustrous)	Dull
Conductivity	Good conductors (of heat & electricity)	Poor conductors (of heat & electricity)
Malleability	Malleable (can be beaten into sheets)	Brittle (cannot be beaten into sheets)
Sonority	Sonorous (produce sound)	Non-sonorous (do not produce sound)
State (at room temp.)	Mostly solids (except mercury)	Solids / Liquids / Gases (e.g., carbon, oxygen)

5. WHERE ARE THEY USED?

Ravi discovers that metals and non-metals are used in different ways in our daily life.

USES OF METALS		
 Copper wires (electricity)	 Iron bridge (construction)	 Gold jewellery (decoration)
USES OF NON-METALS		
 Oxygen (breathing)	 Graphite (pencil lead) (writing)	 Rubber (tyres)

6. SIMPLE TESTS TO IDENTIFY

Ravi learns some simple tests to distinguish metals and non-metals.

	MAGNET TEST Metals like iron, nickel and cobalt are attracted to a magnet. Non-metals are not.
	CONDUCTIVITY TEST Metals conduct electricity and make the bulb glow. Non-metals do not.
	MALLEABILITY TEST Metals can be beaten into thin sheets. Non-metals cannot.

7. COMPARISON AT A GLANCE



METALS

- Shiny
- Good conductors
- Malleable
- Sonorous
- Mostly solids



NON-METALS

- Dull
- Poor conductors
- Brittle
- Non-sonorous
- Solids / Liquids / Gases

8. REAL LIFE APPLICATIONS



Airplanes (Aluminium)



Coins (Nickel)



Cooking utensils (Stainless steel)



Water bottle (Plastic)

The right material is chosen for the right purpose based on its properties.

9. UNDERSTANDING ACHIEVED



Now Ravi understands how metals and non-metals differ and can identify them in his surroundings using their properties.



Knowledge helps us make smarter choices every day!